

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Section: B

1. TITLE

Design a College Information Portal Webpage

2. AIM

To design a multi-section College Information Portal webpage using HTML5 that presents information about the college, its courses, students and registration facilities.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 document.
- To apply text formatting tags and organize content using headings, paragraphs and lists.
- To create hyperlinks, tables and insert images with alternate text.
- To design a student registration form using various form controls.
- To use HTML5 semantic elements for a well-structured layout.
- To embed audio and video using HTML5 media elements.
- To use HTML5 input types such as Email, Date, Number, Color and Range.

4. THEORY

HTML5 is the latest version of the Hyper Text Markup Language used to structure content on the web. It introduces semantic elements such as <header>, <nav>, <section>, <article> and <footer> that describe the meaning of the content they contain, making pages easier to read and maintain.

Text formatting tags such as , <i>, <u>, <sup> and <sub> are used to change how text is displayed. Lists are created using for ordered lists and for unordered lists. The <table> element, along with <tr>, <th> and <td>, is used to display tabular data such as student details.

HTML5 forms use elements like <input>, <select> and <textarea> to collect user data. New input types introduced in HTML5 — email, date, number, color and range — provide built-in validation and specialized controls in the browser. HTML5 also introduced native <audio> and <video> elements to embed media without external plugins.

5. ALGORITHM / PROCEDURE

- 1 Create a new HTML file and declare the HTML5 document type.
- 2 Add the <head> section with a page title.
- 3 Build the page structure using semantic tags — header, nav, main, section, article and footer.

- 4 Add headings and paragraphs describing the college, using text formatting tags.
- 5 Create one ordered list (courses) and one unordered list (facilities).
- 6 Add hyperlinks to the college website and to Google.
- 7 Insert the college logo and a relevant campus image with alternate text.
- 8 Create a table displaying student Roll No, Name, Branch and Email.
- 9 Design a registration form with text, email, password, radio, checkbox, dropdown and textarea fields.
- 10 Add HTML5 input types — Date, Number, Color and Range — inside the form.
- 11 Embed an audio player and a video player using the audio and video elements.
- 12 Save the file and open it in a web browser.
- 13 Verify that every section renders and functions as expected.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">

  <!-- Makes the webpage responsive on mobile devices -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>College Information Portal</title>
</head>

<body>

  <!-- Header section -->
  <header>
    <h1>College Information Portal</h1>

    <p>
      <b>Welcome to our college portal. </b>This webpage provides
      information about the college, students, courses, and
      registration facilities.
    </p>
  </header>

  <!-- Navigation section -->
  <nav>
    <a href="#about">About College</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#registration">Registration</a> |
    <a href="#multimedia">Multimedia</a>
  </nav>

  <hr>

  <!-- Main content -->
  <main>

    <!-- About college section -->
    <section id="about">

      <h2>About Our College</h2>

      <article>
        <p>
          Our college provides <strong>quality education</strong>
          and encourages students to develop
          <em>technical and communication skills</em>.
        </p>
      </article>
    </section>
  </main>
</body>
</html>
```

```

    </p>

    <p>
        The college offers <u>modern learning facilities</u>
        and supports innovation and research.
    </p>

    <p>
        H<sub>2</sub>O is the chemical formula for water.
        The mathematical expression is x<sup>2</sup>.
    </p>
</article>

<h3>Useful Links</h3>

<!-- Hyperlinks -->
<a href="https://www.anits.org" target="_blank">
    Visit College Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
    Open Google
</a>

</section>

<hr>

<!-- Lists section -->
<section id="courses">

    <h2>Courses Offered</h2>

    <!-- Ordered list -->
    <h3>Popular Courses</h3>

    <ol>
        <li>Computer Science and Engineering</li>
        <li>Artificial Intelligence and Machine Learning</li>
        <li>Electronics and Communication Engineering</li>
        <li>Mechanical Engineering</li>
    </ol>

    <!-- Unordered list -->
    <h3>College Facilities</h3>

    <ul>
        <li>Central Library</li>
        <li>Computer Laboratories</li>
        <li>Sports Ground</li>
        <li>College Canteen</li>
    </ul>

</section>

<hr>

<!-- Images section -->
<section>

    <h2>College Images</h2>

    <!-- College logo -->
    
    <!-- Relevant college image -->

```

```

</section>

<hr>

<!-- Student table -->
<section id="students">

    <h2>Student Details</h2>

    <table border="1" cellpadding="10">

        <!-- Table heading -->
        <caption>
            <strong>Student Information</strong>
        </caption>

        <thead>
            <tr>
                <th>Roll No</th>
                <th>Name</th>
                <th>Branch</th>
                <th>Email</th>
            </tr>
        </thead>

        <!-- Table data -->
        <tbody>

            <tr>
                <td>101</td>
                <td>Anusha</td>
                <td>CSE</td>
                <td>anusha@example.com</td>
            </tr>

            <tr>
                <td>102</td>
                <td>Rahul</td>
                <td>AIML</td>
                <td>rahul@example.com</td>
            </tr>

            <tr>
                <td>103</td>
                <td>Priya</td>
                <td>ECE</td>
                <td>priya@example.com</td>
            </tr>

        </tbody>

    </table>

</section>

<hr>

<!-- Student registration form -->
<section id="registration">

    <h2>Student Registration Form</h2>

    <form>

        <!-- Name -->

```

```
<label for="name">Student Name:</label>

<input
  type="text"
  id="name"
  name="name"
  placeholder="Enter your name"
  required
>

<br><br>

<!-- Email -->
<label for="email">Email:</label>

<input
  type="email"
  id="email"
  name="email"
  placeholder="Enter your email"
  required
>

<br><br>

<!-- Password -->
<label for="password">Password:</label>

<input
  type="password"
  id="password"
  name="password"
  required
>

<br><br>

<!-- Gender radio buttons -->
<p><strong>Gender:</strong></p>

<input
  type="radio"
  id="male"
  name="gender"
  value="Male"
>

<label for="male">Male</label>

<input
  type="radio"
  id="female"
  name="gender"
  value="Female"
>

<label for="female">Female</label>

<input
  type="radio"
  id="other"
  name="gender"
  value="Other"
>

<label for="other">Other</label>

<br><br>

<!-- Hobbies checkboxes -->
<p><strong>Hobbies:</strong></p>
<input
```

```
        type="checkbox"
        id="reading"
        name="hobbies"
        value="Reading"
    >

<label for="reading">Reading</label>

<input
    type="checkbox"
    id="sports"
    name="hobbies"
    value="Sports"
>

<label for="sports">Sports</label>

<input
    type="checkbox"
    id="music"
    name="hobbies"
    value="Music"
>

<label for="music">Music</label>

<br><br>

<!-- Branch dropdown -->
<label for="branch">Select Branch:</label>

<select id="branch" name="branch">

    <option value="">-- Select Branch --</option>

    <option value="CSE">
        Computer Science and Engineering
    </option>

    <option value="AIML">
        Artificial Intelligence and Machine Learning
    </option>

    <option value="ECE">
        Electronics and Communication Engineering
    </option>

    <option value="MECH">
        Mechanical Engineering
    </option>

</select>

<br><br>

<!-- Date input -->
<label for="dob">Date of Birth:</label>

<input
    type="date"
    id="dob"
    name="dob"
>

<br><br>

<!-- Number input -->
<label for="age">Age:</label>

<input
    type="number"
    id="age"
```

```

        name="age"
        min="16"
        max="100"
    >

    <br><br>

    <!-- Color input -->
    <label for="colour">Favourite Colour:</label>

    <input
        type="color"
        id="colour"
        name="colour"
    >

    <br><br>

    <!-- Range input -->
    <label for="rating">
        Rate the College:
    </label>

    <input
        type="range"
        id="rating"
        name="rating"
        min="1"
        max="10"
        value="5"
    >

    <br><br>

    <!-- Textarea -->
    <label for="comments">
        Comments:
    </label>

    <br>

    <textarea
        id="comments"
        name="comments"
        rows="5"
        cols="40"
        placeholder="Enter your comments"
    ></textarea>

    <br><br>

    <!-- Submit and reset buttons -->
    <input
        type="submit"
        value="Register"
    >

    <input
        type="reset"
        value="Clear Form"
    >

    </form>

</section>

<hr>

<!-- Audio and video -->
<section id="multimedia">

    <h2>College Multimedia</h2>

```

```
<h3>College Audio</h3>

<audio controls>

  <source
    src="college-audio.mp3"
    type="audio/mpeg"
  >

  Your browser does not support audio.

</audio>

<h3>College Video</h3>

<video
  width="500"
  controls
>

  <source
    src="college-video.mp4"
    type="video/mp4"
  >

  Your browser does not support video.

</video>

</section>

</main>

<hr>

<!-- Footer section -->
<footer>

  <p>
    &copy; 2026 College Information Portal.
    All rights reserved.
  </p>

</footer>

</body>

</html>
```


7. CODE EXPLANATION

Code Explanation

Code	Explanation
<header>, <nav>, <main>, <section>, <article>, <footer>	HTML5 semantic elements used to structure the page into meaningful regions.
, , , <u>, <sup>, <sub>	Format text as bold, strongly emphasized, italic, underlined, superscript and subscript.
/	Display the courses list (ordered) and facilities list (unordered).
	Creates hyperlinks to the college website and Google.
	Displays the college logo and campus image; alt text shows if the image fails to load.
<table>, <caption>, <thead>, <tbody>, <tr>, <th>, <td>	Builds the student details table with Roll No, Name, Branch and Email.
<form>, <input>, <select>, <textarea>	Collects student registration data through various form controls.
type="radio" / "checkbox"	Lets the user select Gender (one option) and Hobbies (multiple options).
type="date" / "number" / "color" / "range"	HTML5 input types providing built-in pickers and validation in the browser.
<audio controls> / <video controls>	Embeds a playable audio element and a playable video element with default browser controls.

8. TEST CASES AND EXECUTION DATA

The webpage was opened in a browser and each section was verified against its expected behaviour.

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-01	Load the page in browser	Header, nav bar, about section and lists display correctly	Displayed correctly	Pass
TC-02	Click "College Website" and "Google" links	Links open the respective pages in a new tab	Opened correctly	Pass
TC-03	Check logo and campus image	Both images render with correct alternate text	Rendered correctly	Pass
TC-04	Check student details table	Roll No, Name, Branch and Email shown in a grid	Displayed correctly	Pass
TC-05	Submit registration form with Name field empty	Browser should require the Name field	Browser blocked submission and asked to fill Name	Pass
TC-06	Select Gender and one or more Hobbies	Only one Gender selectable; multiple Hobbies selectable	Behaved correctly	Pass
TC-07	Pick a Date, Number, Color and Range value	Native browser pickers open for each input type	Pickers opened as expected	Pass
TC-08	Play the Audio and Video elements	Media controls (play/pause) are visible and functional	Controls displayed and functional	Pass

9. OUTPUT

Output 1 — Header, About Section, Text Formatting, Lists, Hyperlinks, Facilities List, College Images, Student Table, Registration Form, Comments, Buttons, Media Elements & Footer

College Information Portal

Welcome to our college portal. This webpage provides information about the college, students, courses, and registration facilities.

[About College](#) | [Courses](#) | [Students](#) | [Registration](#) | [Multimedia](#)

About Our College

Our college provides **quality education** and encourages students to develop *technical and communication skills*.

The college offers modern learning facilities and supports innovation and research.

H₂O is the chemical formula for water. The mathematical expression is x².

Useful Links

[Visit College Website](#)

[Open Google](#)

Courses Offered

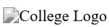
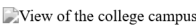
Popular Courses

1. Computer Science and Engineering
2. Artificial Intelligence and Machine Learning
3. Electronics and Communication Engineering
4. Mechanical Engineering

College Facilities

- Central Library
- Computer Laboratories
- Sports Ground
- College Canteen

College Images

Student Details

Student Information			
Roll No	Name	Branch	Email
101	Anusha	CSE	anusha@example.com
102	Rahul	AIML	rahul@example.com
103	Priya	ECE	priya@example.com

Student Registration Form

Student Name:

Email:

Password:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Music

Select Branch:

Date of Birth: 

Age:

Favourite Colour:

Rate the College:

Comments:

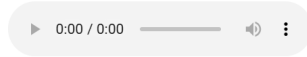
Enter your comments

Register

Clear Form

College Multimedia

College Audio



College Video



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10. RESULT

Thus, a College Information Portal webpage was successfully designed using **HTML5**. Text formatting, lists, hyperlinks, tables, images, a registration form, semantic elements, HTML5 input types and embedded media were implemented, and the output was verified against the expected results for all test cases.